

Quiz 3: Process Scheduling

Matric: **227154** Date: 11.11.2025

1. Given the following processes, show the execution pattern for one cycle, and find out the turnaround time and waiting time for each process, using the stated scheduling policies.

Process	Arrival Time	Processing Time	Priority
A	0	6	3
B	0	4	1
C	0	2	5
D	0	5	4
E	0	3	2

The order of process arrival: A, followed by B, C, D, and E.

First-Come-First-Served (FCFS)

[3M]

[illegible]

Round Robin (quantum time (q) = 3)

[3M]

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	Turnaround Time	Waiting time
A	-	-	-												-	-	-						17	11
B				-	-	-												-					18	14
C							-	-															8	6
D									-	-	-									-	-		20	15
E												-	-	-									14	11
Average																							15.4	11.4

SJF (Shortest Job First) / Shortest Process Next (SPN)

[3M]

[illegible]

Priority

[3M]

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	Turnaround Time	Waiting time
A								-	-	-	-	-	-										13	7
B	-	-	-	-																			4	0
C																			-	-			20	18
D														-	-	-	-	-					18	13
E					-	-	-																7	4
Average																							12.4	8.4

- | Process Name | Arrival Time | Processing Time | Priority |
|--------------|--------------|-----------------|----------|
| A | 0 | 6 | 3 |
| B | 1 | 4 | 1 |
| C | 3 | 2 | 5 |
| D | 5 | 5 | 4 |
| E | 7 | 3 | 2 |

[3M]

[3M]

[3M]

[illegible]

Preemptive SJF (Shortest Job First) / Shortest Remaining Time (SRT)

[3M]

[illegible]

Non-preemptive Priority

[3M]

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	Turnaround Time	Waiting time
A	-	-	-	-	-	-																	6	0
B							-	-	-	-													9	5
C																			-	-			17	15
D														-	-	-	-	-					13	8
E											-	-	-										6	3
Average																							10.2	6.2

Preemptive Priority

[3M]

[illegible]